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Gini Impurity

Introduction:

A decision tree decides where to split by measuring how mixed a group of patient is. Gini impurity gives this measure : 0 means the group is pure (all sick or all healthy), 0.5 means it is a 50-50 mix. At the root, the tree tries every feature, calculates the weighted Gini of the groups it would create, and picks the feature with the lgrgest Gini gain.

Dataset: Heart Disease dataset (first 10 patients)

features: sex, cp, exang

Target: Target = 1 (heart disease present)

Target = 0 (healthy)

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Patients	Sex	cp	exang	Target
1	Male	typical angina	False	0
2	Male	asymptomatic	True	1
3	Male	asymptomatic	True	1
4	Male	non-anginal	False	0
5	Female	atypical angina	False	0
6	Male	atypical anging	False	0
7	Female	asymptomatic	False	1
8	Female	asymptomatic	True	0
9	Male	asymptomatic	False	1
10	Male	asymptomatic	True	1

$$\text{Gini(parent)} = 1 - P(\text{Yes})^2 - P(\text{No})^2$$

$$= 1 - \left(\frac{5}{10}\right)^2 - \left(\frac{5}{10}\right)^2$$

$$\text{Gini(parent)} = 0.5$$

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Split on Sex

	Size	Target	Gini
Male	7	4	$1-(4/7)^2-(3/7)^2=0.490$
Female	3	1	$1-(1/3)^2-(2/3)^2=0.444$

$$\text{Weighted Gini} = \frac{\text{No. of Male} \times \text{Gini of Male} + \text{No. of Female} \times \text{Gini of Female}}{\text{Total}}$$

$$= \frac{7 \times 0.490 + 3 \times 0.444}{10}$$

Weighted Gini = 0.476

Gini gain (Sex) = Gini (parent) − weighted Gini

= 0.5 − 0.476

= 0.024

Split on exang

	Size	Target	Gini
True	4	3	$1-(3/4)^2-(1/4)^2=0.375$
False	6	2	$1-(2/6)^2-(4/6)^2=0.444$

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Weighted Gini = No. of True x Gini of True + No. of False x Gini of False / Total

$= \frac{4 \times 0.375 + 6 \times 0.444}{10}$

Weighted Gini = 0.417

Gini gain (exang) = 0.5 - 0.417

= 0.083

Split on cp

	Size	Target	Gini
typical angina	1	0	$1-(0)^2-(1/1)^2=0$
atypical angina	2	0	$1-(0)^2-(2/2)^2=0$
non anginal	1	0	$1-(0)^2-(1/1)^2=0$
asymptomatic	6	5	$1-(5/6)^2-(1/6)^2=0.278$

Weighted Gini = No. of typical angina x Gini of typical angina + No. of atypical x Gini of atypical + No. of non anginal x Gini of non anginal + No. of asymptomatic x Gini of asymptomatic / Total

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Weighted Gini = $\frac{1 \times 0 + 2 \times 0 + 1 \times 0 + 6 \times 0.278}{10}$

Weighted Gini = 0.167

Gini gain (cp) = 0.5 − 0.167

= 0.333

Gini gain (Sex) = 0.024

Gini gain (exang) = 0.083

Gini gain (cp) = 0.333

cp has the largest Gini gain (0.333), so the tree would choose it as the root split.

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